

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Draft translation - not yet human-reviewed / مسوده ترجمه

Operating System Concepts

Unit 2 named the operating system as the software in charge. This unit opens the door of that office and looks inside: what the operating system actually does all day, how it keeps your files in order, how it shares one processor among many programs, how it moves data to and from devices, and what its innermost part, the kernel, is responsible for. None of this needs programming. It needs the ability to name what is happening when a computer works - which is exactly what lets a teacher manage a lab confidently.

Unit learning outcomes

By the end of this unit you should be able to:

- Describe the **functions** of an operating system and name its main **types**.
- Explain **file management** (names, folders, paths, metadata) and **process management** (what a process is and its states).
- Explain how the operating system manages **input and output** between applications and devices.
- Describe the **kernel** and its functions at the centre of the system.

Prerequisite knowledge

Unit 2, especially Topic 2.3 (system software and application software) and the boot sequence.

In this unit

1. [Topic 3.1 - What an operating system does](#) - its functions and its types.
2. [Topic 3.2 - Files and processes](#) - file management and process management.
3. [Topic 3.3 - Managing input and output](#) - how data reaches devices and back.
4. [Topic 3.4 - The kernel: the core of the system](#) - the innermost layer and its jobs.

How to use this unit

The four topics run from the outside in: what the OS does, then its two great everyday jobs (files and processes), then the device traffic it directs, and finally the core that does the directing. Wherever you sit at a computer, try what each topic describes - open the file list, watch the running-programs view, plug in a USB drive and watch what the OS does. The concepts become concrete the moment you can point at them.

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